

Lab Assignment 6

Demonstrate the use of JSP directives (page, include) and JSP actions (<jsp:include>,<jsp:forward>).

Aim

To demonstrate the use of JSP directives (page, include) and JSP action tags (<jsp:include>, <jsp:forward>) in web applications.

Learning Objectives

1. Understand JSP page directive.
2. Learn static inclusion using include directive.
3. Understand dynamic inclusion using <jsp:include>.
4. Differentiate between static and dynamic inclusion.
5. Implement request forwarding using <jsp:forward>.

Software Requirements

- Java JDK
- Apache Tomcat
- IDE (Eclipse/IntelliJ)
- Web Browser

6.1: Page Directive Demonstration

Create a JSP page to demonstrate the use of the page directive. The page should import the `java.util.Date` class using the page directive and display the current date and time on the browser. Also ensure that the content type of the page is properly set using the directive.

Code:

Date.html

```
<%@ page import="java.util.Date" contentType="text/html; charset=UTF-8" %>
<html>
<body>
<h2>Current Date and Time</h2>
<%
    Date d = new Date();
    out.println(d);
%>
</body>
</html>
```

Output:

Current Date and Time

Mon Mar 23 22:23:03 IST 2026

6.2.: Include Directive (Static Include)

Develop a JSP application to demonstrate the use of the include directive. Create a separate JSP file named header . jsp that contains a heading or title of a website. Then create another JSP page named home . jsp and include the header . jsp file using the include directive. Display some additional content below the included header to show how reusable components can be used

Code:

Home.jsp

```
<%@ include file="Date.jsp" %>
<h2>Welcome to Home Page</h2>
<p>This is static include example.</p>
```

Output:

Current Date and Time

Mon Mar 23 22:26:00 IST 2026

Welcome to Home Page

This is static include example.

6.3.: JSP Action Tag <jsp:include>

Create a JSP application to demonstrate dynamic inclusion using the <jsp:include> action tag. Design a JSP page named menu . jsp that contains a list of links such as Home, About, and Contact. Then create another page named main . jsp and include menu . jsp dynamically using the <jsp:include> tag. Display a welcome message on the main page along with the included menu.

Code:

Menu.jsp

```
<ul>
  <li>Home</li>
  <li>About</li>
  <li>Contact</li>
</ul>
```

<hr>

Main.jsp

```
<html>
<body>
<h2>Welcome to Main Page</h2>
<jsp:include page="menu.jsp" />
<p>This is dynamic include example.</p>
</body>
</html>
```

Output:

Welcome to Main Page

- Home
- About
- Contact

This is dynamic include example.

6.4: Difference Between Include Directive and <jsp:include>

Develop a JSP program to compare the include directive and the <jsp:include> action tag. Create a JSP file that displays some dynamic content (for example, current time or a value passed through request parameters). Include this file in one JSP page using the include directive and in another JSP page using the <jsp:include> tag. Observe the output in both cases and explain the difference between static and dynamic inclusion.

Code:

Dynamic.jsp

```
<%
    out.println("Current Time: " + new java.util.Date() + "<br>");

    String name = request.getParameter("name");
    if(name != null){
        out.println("Hello " + name);
    }
%>
```

staticInculde.jsp

```
<%@ include file="dynamic.jsp" %>

<h3>This is Static Include Page</h3>
```

dynaicInclude.jsp

```
<html>
<body>

<h3>This is Dynamic Include Page</h3>

<jsp:include page="dynamic.jsp">
    <jsp:param name="name" value="Shriyash" />
</jsp:include>

</body>
</html>
```

Output:

Current Time: Mon Mar 23 22:31:49 IST 2026

This is Static Include Page

This is Dynamic Include Page

Current Time: Mon Mar 23 22:31:58 IST 2026
Hello Shriyash

6.5: JSP Action Tag <jsp:forward>

Create a JSP-based application to demonstrate request forwarding using the <jsp:forward> action tag. Design a page named input.jsp that accepts user input such as a name through a form. After submission, forward the request to another page named welcome.jsp using <jsp:forward>. The welcome.jsp page should retrieve the user input and display a personalized welcome message.

Code:

Input.jsp

```
<html>
<body>

<h2>Enter Your Name</h2>

<form action="welcome.jsp" method="post">
  Name: <input type="text" name="username">
  <input type="submit" value="Submit">
</form>

</body>
</html>
```

Welcome.jsp

```
<%
  String name = request.getParameter("username");
%>

<jsp:forward page="display.jsp">
```

```
<jsp:param name="user" value="<%= name %>" />
</jsp:forward>
```

Display.jsp

```
<%
    String user = request.getParameter("user");
%>
```

```
<h2>Welcome, <%= user %>!/</h2>
```

Output:

Enter Your Name

Name:

Welcome, Shriyash!

Conclusion (JSP)

In this experiment, we studied and implemented various features of Java Server Pages (JSP), including directives and action tags. The page directive was used to configure JSP pages and import Java classes, while the include directive enabled static inclusion of reusable components.

We also explored JSP action tags such as `<jsp:include>` for dynamic content inclusion and `<jsp:forward>` for request forwarding between pages. These features help in building modular, reusable, and efficient web applications.

Overall, JSP simplifies web development by combining HTML with Java code, allowing dynamic content generation and better management of web applications.